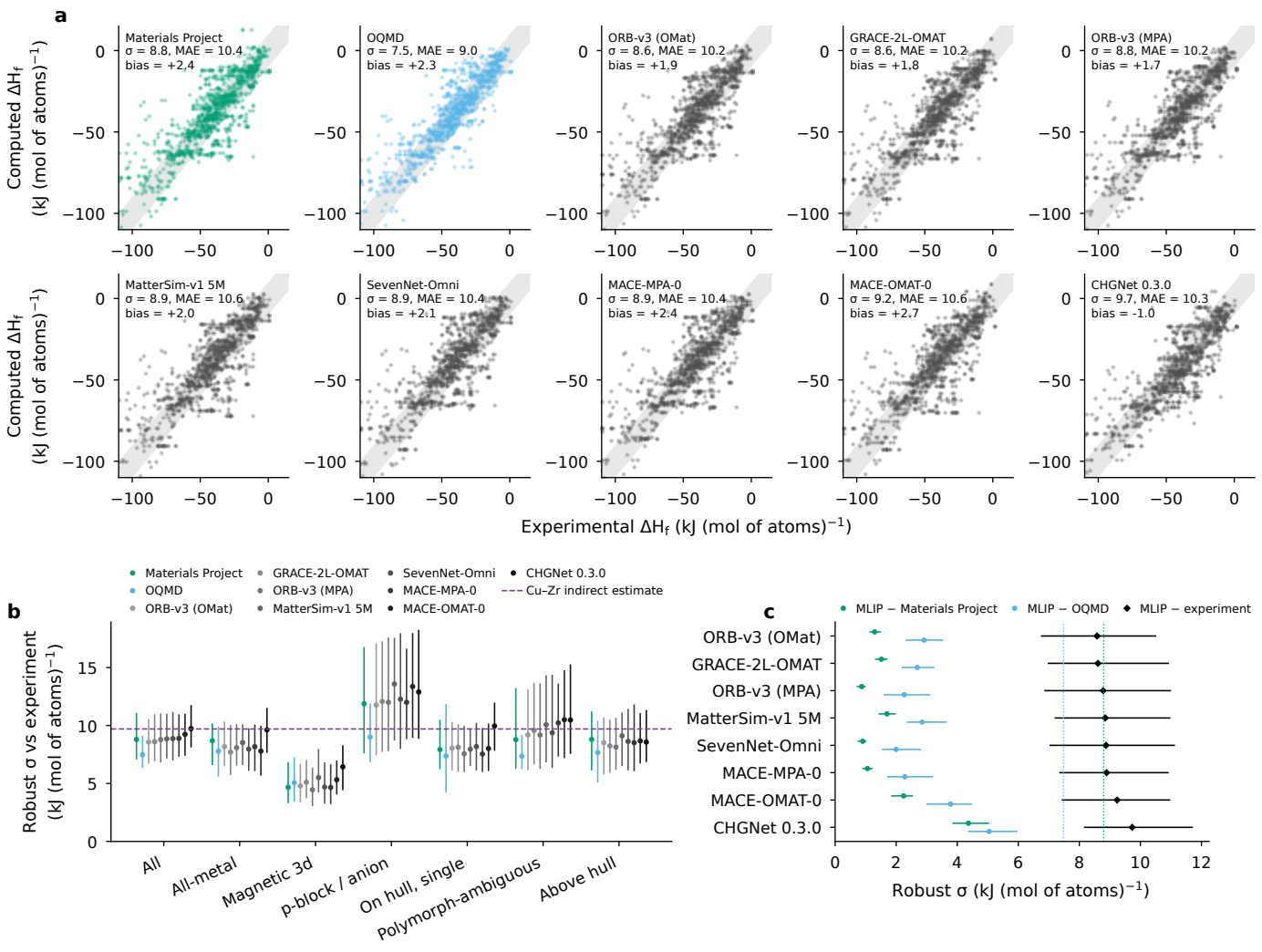

a
Materials Project
σ = 8.8, MAE = 10.4
bias = +2.4
OQMD
σ = 7.5, MAE = 9.0
bias = +2.3
ORB-v3 (OMat)
σ = 8.6, MAE = 10.2
bias = +1.9
GRACE-2L-OMAT
σ = 8.6, MAE = 10.2
bias = +1.8
ORB-v3 (MPA)
σ = 8.8, MAE = 10.2
bias = +1.7
MatterSim-v1 5M
σ = 8.9, MAE = 10.6
bias = +2.0
SevenNet-Omni
σ = 8.9, MAE = 10.4
bias = +2.1
MACE-MPA-0
σ = 8.9, MAE = 10.4
bias = +2.4
MACE-OMAT-0
σ = 9.2, MAE = 10.6
bias = +2.7
CHGNet 0.3.0
σ = 9.7, MAE = 10.3
bias = -1.0
Computed ΔHf (kJ (mol of atoms)−1)
Experimental ΔHf (kJ (mol of atoms)−1)
b
Materials Project
OQMD
ORB-v3 (OMat)
GRACE-2L-OMAT
ORB-v3 (MPA)
MatterSim-v1 5M
SevenNet-Omni
MACE-MPA-0
MACE-OMAT-0
CHGNet 0.3.0
Cu–Zr indirect estimate
Robust σ vs experiment (kJ (mol of atoms)−1)
All
All-metal
Magnetic 3d
p-block / anion
On hull, single
Polymorph-ambiguous
Above hull
c
MLIP − Materials Project
MLIP − OQMD
MLIP − experiment
ORB-v3 (OMat)
GRACE-2L-OMAT
ORB-v3 (MPA)
MatterSim-v1 5M
SevenNet-Omni
MACE-MPA-0
MACE-OMAT-0
CHGNet 0.3.0
Robust σ (kJ (mol of atoms)−1)